

DEPARTMENT OF MATHEMATICS
MAT122 INTRODUCTORY MATHEMATICS II

WEEK 4: LECTURE NOTE

DIFFERENTIATION continued.....

Product and Quotient Rules and Higher-Order Derivatives

The proof of the following theorem follows from properties of limit.

Theorem 1 (The Product Rule). *The product of two differentiable functions f and g is itself differentiable. Moreover, the derivative of fg is the first function times the derivative of the second, plus the second function times the derivative of the first; that is,*

$$\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x). \quad [\text{Product Rule}]$$

Example 2. Find the derivative of $h(x) = (3x - 2x^2)(5 + 4x)$.

Solution.

$$\begin{aligned} h'(x) &= (5 + 4x) \frac{d}{dx}[3x - 2x^2] + (3x - 2x^2) \frac{d}{dx}[5 + 4x] && (\text{Product Rule}) \\ &= (5 + 4x)[3 - 4x] + (3x - 2x^2)4 && (\text{Constant Multiple, Sum and Difference Rules}) \\ &= 15 - 8x - 16x^2 + 12x - 8x^2 && (\text{Expand}) \\ &= 15 + 4x - 24x^2 && (\text{Simplify}) \end{aligned}$$

Example 3. Find the derivative of $y = 3x^2 \sin x$.

Solution. Reason out each of the steps.

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}[3x^2 \sin x] \\ &= 3x^2 \frac{d}{dx}[\sin x] + \sin x \frac{d}{dx}[3x^2] \\ &= 3x^2 \cos x + 6x \sin x \end{aligned}$$

Example 4. Find the derivative of $y = 2x \cos x - 2 \sin x$.

Solution. Reason out each of the steps.

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}[2x \cos x - 2 \sin x] \\ &= \frac{d}{dx}[2x \cos x] - \frac{d}{dx}[2 \sin x] \\ &= 2x \frac{d}{dx}[\cos x] + \cos x \frac{d}{dx}[2x] - 2 \frac{d}{dx}[\sin x] \\ &= -2x \sin x + 2 \cos x - 2 \cos x \\ &= -2x \sin x \end{aligned}$$

Theorem 5 (The Quotient Rule). *The quotient $\frac{f}{g}$ of two differentiable functions f and g is itself differentiable at all values of x for which $g(x) \neq 0$. Moreover, the derivative of $\frac{f}{g}$ is*

given by the denominator times the derivative of the numerator minus the numerator times the derivative of the denominator, all divided by the square of the denominator; that is,

$$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}, \quad g(x) \neq 0. \quad [\text{The Quotient Rule}]$$

Remark 6. The Quotient Rule can be extended to any finite number of functions.

Example 7. Find the derivative of $y = \frac{5x-2}{x^2+1}$.

Solution. Reason out each of the steps.

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}\left[\frac{5x-2}{x^2+1}\right] \\ &= \frac{(x^2+1)\frac{d}{dx}[5x-2] - (5x-2)\frac{d}{dx}[x^2+1]}{(x^2+1)^2} \\ &= \frac{5(x^2+1) - 2x(5x-2)}{(x^2+1)^2} \\ &= \frac{-5x^2 + 4x + 5}{(x^2+1)^2} \end{aligned}$$

Example 8. Find an equation of the tangent line to the graph of $f(x) = \frac{3-\frac{1}{x}}{x+5}$ at the point $P(-1, 1)$.

Solution. Begin by rewriting the function

$$f(x) = \frac{x(3 - \frac{1}{x})}{(x)x + 5} = \frac{3x - 1}{x^2 + 5x}$$

Applying the Quotient Rule for differentiation, we obtain

$$\begin{aligned} f'(x) &= \frac{3(x^2 + 5x) - (3x - 1)(2x + 5)}{(x^2 + 5x)^2} \\ &= \frac{[3x^2 + 15x] - [6x^2 + 13x - 5]}{(x^2 + 5x)^2} \\ &= \frac{-3x^2 + 2x + 5}{(x^2 + 5x)^2} \end{aligned}$$

Therefore, the slope m of the tangent line to the graph of $f(x)$ at $(-1, 1)$ is

$$m = f'(-1) = 0.$$

Then, using the point-slope form of the equation of a line, you can determine that the equation of the tangent line at $(-1, 1)$ is the horizontal line $y = 1$.

Not every quotient needs to be differentiated by the Quotient Rule. For instance, each quotient in the next example can be considered as the product of a constant times a function. In such cases, it is more convenient to use the Constant Multiple Rule.

Example 9. Using the Constant Multiple Rule

<i>Original Function</i>	<i>Rewrite</i>	<i>Differentiate</i>	<i>Simplify</i>
$y = \frac{x^2 + 3x}{6}$	$y = \frac{1}{6}(x^2 + 3x)$	$y' = \frac{1}{6}(2x + 3)$	$y' = \frac{1}{3}x + \frac{1}{2}$
$y = \frac{5x^4}{8}$	$y = \frac{5}{8}x^4$	$y' = \frac{5}{8}[4x^3]$	$y' = \frac{5}{2}x^3$
$y = \frac{-3(3x - 2x^2)}{7x}$	$y = \frac{-3}{7}(3 - 2x)$	$y' = \frac{-3}{7}(-2)$	$y' = \frac{6}{7}$
$y = \frac{9}{5x^2}$	$y = \frac{9}{5}(x^{-2})$	$\frac{9}{5}(-2x^{-3})$	$\frac{-18}{5x^3}$

Example 10 (Power Rule: Negative Integer Exponents). *Let n be a negative integer. Find*

$$\frac{d}{dx}[x^n].$$

Solution. *Since n is a negative integer, there exists a positive integer k such that $n = -k$. So, by the Quotient Rule,*

$$\begin{aligned}
 \frac{d}{dx}[x^n] &= \frac{d}{dx}[x^{-k}] \\
 &= \frac{d}{dx}\left[\frac{1}{x^k}\right] \\
 &= \frac{0 - \frac{d}{dx}[x^k]}{(x^k)^2} \\
 &= \frac{-kx^{k-1}}{x^{2k}} \\
 &= -kx^{-k-1} \\
 &= nx^{n-1}
 \end{aligned}$$

So, the Power Rule

$$\frac{d}{dx}[x^n] = nx^{n-1} \quad (1)$$

is valid for any integer.

Derivatives of Trigonometric Functions

Recall that we have obtained the derivatives of the sine, the cosine and the tangent functions by using the limit process. Knowing the derivatives of the sine and cosine functions, we can use the Quotient Rule to find the derivatives of the four remaining trigonometric functions. The proofs are left as an exercise.

Theorem 11. *Derivatives of Trigonometric Functions*

$$\frac{d}{dx}[\tan x] = \sec^2 x \quad (2)$$

$$\frac{d}{dx}[\cot x] = -\csc^2 x \quad (3)$$

$$\frac{d}{dx}[\sec x] = \sec x \tan x \quad (4)$$

$$\frac{d}{dx}[\csc x] = -\csc x \cot x \quad (5)$$

Example 12. *Differentiate*

- (a) $y = x - \tan x$
 (b) $y = x \sec x$
 (c) $y = \frac{1 - \cos x}{\sin x}$

Solution.

$$(a) \quad y' = \frac{d}{dx}[x - \tan x] = \frac{d}{dx}[x] - \frac{d}{dx}[\tan x] = 1 - \sec^2 x.$$

$$(b) \quad y' = \frac{d}{dx}[x \sec x] = \sec x \frac{d}{dx}[x] + x \frac{d}{dx}[\sec x] = \sec x + x \sec x \tan x.$$

(c) *We evaluate the limit as follows*

$$\begin{aligned} y' &= \frac{d}{dx} \left[\frac{1 - \cos x}{\sin x} \right] \\ &= \frac{\sin x \frac{d}{dx}[1 - \cos x] - (1 - \cos x) \frac{d}{dx}[\sin x]}{\sin^2 x} \\ &= \frac{\sin x - (1 - \cos x) \cos x}{\sin^2 x} \\ &= \frac{1 - \cos x}{\sin^2 x} \\ &= \csc^2 x - \cot x \csc x. \end{aligned}$$

Higher-Order DerivativesLet $f(x) = x^4 + \cos x$. Then

$$f'(x) = 4x^3 - \sin x$$

is also a function. We call $f' = \frac{d}{dx}[f(x)]$ the **first order derivative** of f . If we differentiate the new function f' , we obtain

$$\frac{d}{dx}[f'(x)] = 12x^2 - \cos x.$$

We call it **second order derivative** of f and denote it by

$$f'' \quad \text{or} \quad \frac{d^2}{dx^2}[f(x)]$$

This leads us to higher order derivatives. The second derivative is an example of a **higher-order derivative**.

Higher-order derivatives are denoted as shown below.

First derivative:	y' ,	$f'(x)$,	$\frac{dy}{dx}$,	$\frac{d}{dx}[f(x)]$,	or $D_x[y]$
Second derivative:	y'' ,	$f''(x)$,	$\frac{d^2 y}{dx^2}$,	$\frac{d^2}{dx^2}[f(x)]$,	or $D_x^2[y]$
Third derivative:	y''' ,	$f'''(x)$,	$\frac{d^3 y}{dx^3}$,	$\frac{d^3}{dx^3}[f(x)]$,	or $D_x^3[y]$
Fourth derivative:	$y^{(4)}$,	$f^{(4)}(x)$,	$\frac{d^4 y}{dx^4}$,	$\frac{d^4}{dx^4}[f(x)]$,	or $D_x^4[y]$
$\vdots$					
n^{th} derivative:	$y^{(n)}$,	$f^{(n)}(x)$,	$\frac{d^n y}{dx^n}$,	$\frac{d^n}{dx^n}[f(x)]$,	or $D_x^n[y]$

Example 13 (Finding the Acceleration Due to Gravity). *Because the moon has no atmosphere, a falling object on the moon encounters no air resistance. In 1971, astronaut David Scott demonstrated that a feather and a hammer fall at the same rate on the moon. The position function for each of these falling objects is*

$$s(t) = -0.81t^2 + 2$$

where s is the height in meters and t is the time in seconds. What is the ratio of Earth's gravitational force to the moon's?

Solution. To find the acceleration, differentiate the position function twice.

$s(t) = -0.81t^2 + 2$	Position function
$s'(t) = -1.62t$	Velocity function
$s''(t) = -1.62$	Acceleration function

So, the acceleration due to gravity on the moon is $-1.62 \frac{m}{s^2}$. Because the acceleration due to gravity on Earth is $-9.8 \frac{m}{s^2}$, the ratio of Earth's gravitational force to the moon's is

$$\frac{\text{Earth's gravitational force}}{\text{Moon's gravitational force}} = \frac{-9.8}{-1.62} \approx 6$$

The Earth's gravitational force is six times that of the Moon's.

Example 14. Find Fifth Order Derivative of the function $f(x) = \ln x + \sin x$ for $x > 0$.

Solution.

$f(x) = \ln x + \sin x$	Original function
$f'(x) = \frac{1}{x} + \cos x$	First Derivative
$f''(x) = -\frac{1!}{x^2} - \sin x$	Second Derivative
$f'''(x) = \frac{2!}{x^3} - \cos x$	Third Derivative
$f^{(4)}(x) = -\frac{3!}{x^4} + \sin x$	Forth Derivative
$f^{(5)}(x) = \frac{4!}{x^4} + \cos x$	Fifth Derivative

Example 15. Find n^{th} Order Derivative of the function $f(x) = \frac{1}{x}$ for $x \neq 0$.

Solution.

$f(x) = \frac{1}{x}$	Original function
$f'(x) = -\frac{1!}{x^2}$	First Derivative
$f''(x) = \frac{2!}{x^3}$	Second Derivative
$f'''(x) = -\frac{3!}{x^4}$	Third Derivative
$f^{(4)}(x) = \frac{4!}{x^5}$	Forth Derivative
$\vdots$	
$f^{(n)}(x) = (-1)^n \frac{n!}{x^{n+1}}$	n^{th} Derivative

The Chain Rule

The Chain Rule deals with differentiation of composite functions. Note that a composite function $f \circ g$ has two parts - an **inner part** and an **outer part**.

Example 16. *Inner and Outer parts of $f \circ g$*

Given function $y = f(g(x))$	Outer function $y = f(u)$	Inner function $u = g(x)$
$y = \sqrt{x^2 + 1}$	$f(u) = \sqrt{u}$	$u = x^2 + 1$
$y = \sin(6x)$	$f(u) = \sin u$	$u = 6x$
$y = (3x + 2)^5$	$f(u) = u^5$	$u = 3x + 2$
$y = (x + \tan x)^2$	$f(u) = u^2$	$u = x + \tan x$
$y = \frac{1}{x+1}$	$f(u) = \frac{1}{u}$	$u = x + 1$
$y = \tan^2 x$	$f(u) = \tan u$	$u = \tan x$
$y = \sqrt{3x^2 - x + 1}$	$f(u) = \sqrt{u}$	$u = 3x^2 - x + 1$

Theorem 17 (THE CHAIN RULE). *If*

$$y = f(u)$$

is a differentiable function of u and

$$u = g(x)$$

is a differentiable function of x , then

$$y = f(g(x))$$

is a differentiable function of x and

$$\boxed{\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}} \quad (6)$$

or, equivalently,

$$\boxed{\frac{d}{dx}[f(g(x))] = f'(g(x)) g'(x)}. \quad (7)$$

Proof. *Let*

$$h(x) = f(g(x)).$$

If

$$g(x + h) = g(x) + k(h),$$

then by the continuity of g at x

$$\lim_{h \rightarrow 0} k(h) = \lim_{h \rightarrow 0} [g(x + h) - g(x)] = 0$$

so that by the continuity of $f \circ g$ at $g(x)$

$$\lim_{h \rightarrow 0} f(g(x + h)) = \lim_{k \rightarrow 0} f(g(x) + k) = f(g(x)). \quad (8)$$

With this fact and the definition of derivative we have

$$\begin{aligned}
 h'(x) &= \frac{d}{dx}[f(g(x))] \\
 &= \lim_{h \rightarrow 0} \frac{f(g(x+h)) - f(g(x))}{h} \\
 &= \lim_{h \rightarrow 0} \left[\frac{f(g(x+h)) - f(g(x))}{h} \right] \left[\frac{g(x+h) - g(x)}{g(x+h) - g(x)} \right] \\
 &= \lim_{h \rightarrow 0} \left[\frac{f(g(x+h)) - f(g(x))}{g(x+h) - g(x)} \right] \left[\frac{g(x+h) - g(x)}{h} \right] \\
 &= \lim_{h \rightarrow 0} \left[\frac{f(g(x+h)) - f(g(x))}{g(x+h) - g(x)} \right] \lim_{h \rightarrow 0} \left[\frac{g(x+h) - g(x)}{h} \right] \\
 &= \lim_{h \rightarrow 0} \left[\frac{f(g(x) + k(h)) - f(g(x))}{k(h)} \right] \lim_{h \rightarrow 0} \left[\frac{g(x+h) - g(x)}{h} \right] \\
 &= \lim_{k \rightarrow 0} \left[\frac{f(g(x) + k) - f(g(x))}{k} \right] g'(x) \\
 &= f'(g(x)) g'(x).
 \end{aligned}$$

The derivative of $y = f(u)$ is the derivative of the outer function f (at the inner function u) times the derivative of the inner function g

Example 18. Use the Chain Rule to find the derivative of $f(x) = (x^2 + 1)^3$.

Solution. For this function, you can consider the inner function to be $u = g(x) = x^2 + 1$ and the outer function to be $y = f(u) = u^3$. By the Chain Rule, you obtain

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = 3u^2 \cdot (2x) = 6x(x^2 + 1)^2$$

The following theorem is a consequence of the Chain Rule.

Theorem 19 (The General Power Rule). If $y = [u(x)]^r$, where u is a differentiable function of x and r is a rational number, then

$$\frac{dy}{dx} = r[u(x)]^{r-1} u'(x) \quad (9)$$

or, equivalently,

$$\frac{d}{dx}[u^r] = ru^{r-1} \frac{du}{dx}. \quad (10)$$

Example 20. Use the General Power Rule to find the derivative of $f(x) = (3x - 2x^2)^3$.

Solution. Let $u = 3x - 2x^2$. Then

$$y = (3x - 2x^2)^3 = u^3$$

and, by the General Power Rule

$$f'(x) = 3[u(x)]^{3-1} u'(x) = 3(3x - 2x^2)^2(3 - 4x) = (9 - 12x)(3x - 2x^2)^2.$$

Example 21. Find all points on the graph of

$$f(x) = \sqrt[3]{(x^2 - 1)^2}$$

for which $f'(x) = 0$ and those for which $f'(x)$ does not exist.

Solution. Begin by rewriting the function as

$$f(x) = \sqrt[3]{(x^2 - 1)^2} = (x^2 - 1)^{\frac{2}{3}}$$

Then, applying the General Power Rule (with $u = x^2 - 1$ produces

$$f'(x) = \frac{2}{3}u^{\frac{2}{3}-1} \cdot \frac{du}{dx} = \frac{2}{3}(x^2 - 1)^{-\frac{1}{3}} \cdot (2x) = \frac{4}{3}x(x^2 - 1)^{-\frac{1}{3}} = \frac{4x}{\sqrt[3]{x^2 - 1}}$$

So, $f'(x) = 0$ when $x = 0$, and $f'(x)$ does not exist when $x = \pm 1$.

Example 22. Differentiate the function $g(t) = \frac{-7}{(2t - 3)^2}$.

Solution. Begin by rewriting the function as

$$g(t) = -7(2t - 3)^{-2}.$$

Then, applying the General Power Rule (with $u = 2t - 3$) produces

$$g'(t) = -7(-2)(2t - 3)^{-2-1}(2) = 28(2t - 3)^{-2-1} = \frac{28}{(2t - 3)^3}$$

Simplifying derivatives using algebra

The following examples demonstrate techniques for simplifying the "raw derivatives" of functions involving products, quotients, and composites algebraically.

Example 23. Find the derivative of

$$f(x) = x^2\sqrt{1 - x^2}.$$

Solution. Rewrite the function as $f(x) = x^2(1 - x^2)^{\frac{1}{2}}$. Then

$$\begin{aligned} f'(x) &= (1 - x^2)^{\frac{1}{2}} \frac{d}{dx}[x^2] + x^2 \frac{d}{dx}[(1 - x^2)^{\frac{1}{2}}] && \text{[Product Rule]} \\ &= 2x(1 - x^2)^{\frac{1}{2}} + x^2 \left[\frac{1}{2}(1 - x^2)^{\frac{1}{2}-1} \frac{d}{dx}[1 - x^2] \right] && \text{[Chain Rule and Power Rule]} \\ &= 2x(1 - x^2)^{\frac{1}{2}} + x^2 \left[\frac{1}{2}[(1 - x^2)^{-\frac{1}{2}}] [-2x] \right] \\ &= 2x(1 - x^2)^{\frac{1}{2}} - x^3(1 - x^2)^{-\frac{1}{2}} \\ &= 2x\sqrt{1 - x^2} - \frac{x^3}{\sqrt{1 - x^2}} \\ &= \frac{2x(1 - x^2) - x^3}{\sqrt{1 - x^2}} \\ &= \frac{x(2 - 3x^2)}{\sqrt{1 - x^2}} \end{aligned}$$

Example 24. Differentiate

$$\begin{aligned} (a) \quad f(x) &= \frac{x}{\sqrt[3]{x^2 + 4}} \\ (b) \quad f(x) &= \left(\frac{3x - 1}{x^2 + 3} \right)^2 \end{aligned}$$

Solution.

(a) Rewrite the function to obtain

$$f(x) = \frac{x}{\sqrt[3]{x^2 + 4}} = \frac{x}{(x^2 + 4)^{\frac{1}{3}}}.$$

Thus we have

$$\begin{aligned} f'(x) &= \frac{(x^2 + 4)^{\frac{1}{3}} \frac{d}{dx}[x] - x \frac{d}{dx}[(x^2 + 4)^{\frac{1}{3}}]}{[(x^2 + 4)^{\frac{1}{3}}]^2} \\ &= \frac{(x^2 + 4)^{\frac{1}{3}} - x \frac{1}{3}(x^2 + 4)^{\frac{1}{3}-1} \frac{d}{dx}[x^2 + 4]}{(x^2 + 4)^{\frac{2}{3}}} \\ &= \frac{(x^2 + 4)^{\frac{1}{3}} - \frac{2x^2}{3}(x^2 + 4)^{-\frac{2}{3}}}{(x^2 + 4)^{\frac{2}{3}}} \\ &= \frac{(x^2 + 4)^{\frac{1}{3}} - \frac{2x^2}{3}(x^2 + 4)^{-\frac{2}{3}}}{(x^2 + 4)^{\frac{2}{3}}} \\ &= \frac{3(x^2 + 4) - 2x^2}{3(x^2 + 4)^{\frac{4}{3}}} \\ &= \frac{x^2 + 12}{3(x^2 + 4)^{\frac{4}{3}}} \end{aligned}$$

QUOTIENT RULE

POWER and CHAIN RULE

(b) Here we simplify the power

$$\begin{aligned} f'(x) &= 2\left(\frac{3x-1}{x^2+3}\right)^{2-1} \frac{d}{dx}\left[\frac{3x-1}{x^2+3}\right] \\ &= 2\left(\frac{3x-1}{x^2+3}\right) \frac{3(x^2+3) - (3x-1)(2x)}{(x^2+3)^2} \\ &= \frac{2(3x-1)(-3x^2+2x+9)}{(x^2+3)^3} \end{aligned}$$

POWER RULE

QUOTIENT RULE

SIMPLIFY